CCC-C-428F
October 28, 1981
SUPERSEDING
Fed. Spec. CCC-C-428E
November 5, 1971

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON; FIRE, WATER, WEATHER, AND MILDEW RESISTANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for fire, water, weather, and mildew resistant cotton duck.

1.2 Classification.

1.2.1 Types and classes. The finished cloth shall be of the following types and classes as specified (see 6.2):

Type I - Cloth finished in Olive Drab 7
Type II - Cloth finished in deck grey
Class 1 - Regular finish
Class 2 - Dry finish

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-C-419	-	Cloth, Duck, Cotton, Unbleached, Plied-Yarns Army and Numbered
CCC-C-443	-	Cloth, Duck, Cotton (Single and Plied Filling Yarns, Flat)
PPP-P-1135	-	Packaging and Packing of Duck Fabrics (Cotton; Synthetic Fiber; Cotton-synthetic Fiber Blends)

Federal Standards:

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods

FSC 8305

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(Single copies of this specification, other Federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Washington, DC; Philadelphia; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

(Copies of Military specifications and standards required by contractors in connection with specific procurement should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is established, the cloth shall match the standard sample for shade and shall be equal to, or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

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3.2 Laboratory report approval. Unless otherwise specified, the prospective contractor shall submit to the contracting officer for approval at the time of submission of bid, a certified copy of a recent laboratory report covering the finished cloth which he proposes to deliver (see 6.2). Unless otherwise specified by the contracting officer, the bidder shall certify that the laboratory report submitted covers cloth which was finished in a mill where the finishing will be performed if the contract is awarded. The laboratory report shall contain test data which demonstrates that the finished cloth which the contractor proposes to deliver has been tested and found to comply with all requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement.

(a) An independent or commercial laboratory report.
(b) The prospective bidder's own laboratory report.
(c) A Government laboratory report from a contract within 6 months of date of submission of bid.

The purpose of the report is to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of the specification.

3.3 Formula approval (see 4.1.2). The prospective contractor or his finishing plant shall submit with their bid to the contracting officer for approval, a list of materials which the prospective contractor or his finishing plant intends to use, stipulating amounts, grades, and the standards of all materials including solvents and manufacturer's name (see 6.2). This list is for the information of the Government only, and is not to be made available to the public. The use of compounds containing mercury in any form is prohibited. The use of proteins and their derivatives, starch, hydrophylic materials, oxidizing oils and oxidizing resins, and resins or compounds to increase hydroscopicity is prohibited unless specifically approved by the contracting officer as part of the formula submitted hereunder. Approval of the qualitative formula submitted shall in no way absolve the contractor from complying with the performance requirements of this specification.

3.4 Materials.

3.4.1 Base cloth. The base cloth shall be as specified by the contracting agency and shall conform to CCC-C-419 or CCC-C-443 (see 6.2), except that the nonfibrous material for CCC-C-443 shall be as specified in 3.4.1.1.

3.4.1.1 Nonfibrous material content. The cloth prior to finishing, shall contain not more than 2.5 percent starch and protein content excluding chloroform-soluble and water-soluble material and not more than 5.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.4.

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3.4.2 Grey-cloth weight. Unless otherwise specified, where reference is made to grey-cloth weight, the minimum acceptable grey-cloth weight as given in the applicable grey-cloth specification shall be used.

3.4.3 Compound. The compound to provide fire, water, weather, and mildew resistance shall be well ground and evenly blended. Compounds containing sulfur may be permitted when specifically indicated by name in formulas approved by the contracting officer. The compound shall be of the same ingredients as contained in the list of materials submitted and approved under 3.3.

3.5 Finished cloth.

3.5.1 Construction. The number of yarns per inch of the finished cloth shall conform to the requirements of the specification covering the grey cloth specified except that a minus tolerance in the filling yarn will be allowed as follows:

Designation	Minus tolerance
Flat ducks (requiring desizing)	Up to 10 percent
Army duck	Up to 5 percent
No. 8 and lighter duck	Up to 5 percent
Heavier than No. 8 duck	Up to 3 percent

3.5.1.1 Shuttleless loom selvages. When a clot is woven on shuttleless looms, the fringed selvages shall be removed from the cloth (not past point of lock-in or into body of cloth).

3.5.2 Finished width. The nominal width of the finished cloth shall be as specified (see 6.2). The minimum acceptable width shall be inclusive of the woven selvage when fly-shuttle looms are used and exclusive of the lock in selvage when shuttleless looms are used. Raw edges shall not be permitted. The following tolerances shall apply:

Numbered duck

Width of duck Inches	Tolerance Inch
40 or less	-1/2 +1/4
Over 40 to 51 inclusive	-1/2 +3/8
Over 51 to 81 inclusive	-3/4 +1/2
Over 81	-7/8 +5/8

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Army duck

Width of duck Inches	Tolerance Inch
40 or less	-1/4 +1/2
Over 40 to 51 inclusive	-1/4 +5/8
Over 51 to 81 inclusive	-3/8 +3/4
Over 81	-1/2 +7/8

3.5.3 Breaking strength. The breaking strength of the finished cloth in both warp and filling directions shall be not less than that of the corresponding specification requirements for the grey cloth.

3.5.4 Weight. The weight of the finished cloth shall be the actual weight of the grey cloth plus an allowance for the treatment in accordance with table I.

TABLE I

Grey weight (ounces per sq yd)	Allowable maximum increase in weight due to treatment based on actual grey-cloth weight (percent)
12 and over	50
Under 12	65

3.5.5 Color. Unless otherwise specified, the color shall be as specified for the applicable type and shall match the standard sample when furnished by the procuring activity (see 6.4). The color shall be obtained with carbon black or lamp black, inorganic or phthalocyanine pigments, or such other coloring materials and shall not be appreciably affected by exposure to weathering when tested as specified in 4.4.

3.5.5.1 Colorfastness. When a standard sample is established, the colored and finished cloth shall show fastness to accelerated weathering equal to or better than the standard sample when tested as specified in 4.4. When a standard sample is not available or is not referenced as a standard for colorfastness, the colored and finished cloth shall show "good" colorfastness to accelerated weathering when tested as specified in 4.4.

3.5.5.2 Matching. The color of the finished cloth shall match the standard shade under artificial daylight having a color temperature of 7000 $\pm$ 500 kelvin and shall be a good approximation of the standard sample under incandescent lamplight at 2850 $\pm$ 100 kelvin.

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3.5.6 Crocking.

3.5.6.1 Class 1. Cloths with a regular finish shall show no more crocking on dry white cotton cloth than Munsell value of 5.0 when tested as specified in 4.4.

3.5.6.2 Class 2. Cloths with a dry finish for upholstery purposes shall show no more crocking on dry white cotton cloth than Munsell value of 7.0 when tested as specified in 4.4.

3.5.7 Flexibility. The finished cloth shall show a bending moment of not more than that specified in table II when tested as specified in 4.4.

TABLE II. Flexibility

CCC-C-419 cloth designations 1/	Initial bending moment inch-lbs maximum	After heating at 200°-205°F (bending moment inch-lbs) maximum	At 0° ± 5°F (bending moment inch-lbs) maximum
Numbered			
2	0.16	0.400	0.400
4	0.12	0.300	0.300
6	0.06	0.180	0.180
8	0.05	0.125	0.125
10	0.03	0.075	0.075
12	0.02	0.050	0.050
Army ducks			
8.25	0.013	0.032	0.032
9.85	0.013	0.032	0.032
12.29	0.013	0.032	0.032

1/ For all other cloths, the flexibility requirements shall be as specified (see 6.2).

3.5.8 Water resistance. The finished cloth shall have a maximum leakage of 50 mL at the height indicated in table III when tested by the water permeability test specified in 4.4. The cloth shall conform to the same maximum leakage requirements after the accelerated weathering test specified in 4.4.

3.5.8.1 Applicability. The requirements for resistance to water permeability shall be applicable to all ducks of CCC-C-419 only.

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3.5.8.2 Retest. When the finished cloth fails to comply with 3.5.8, a retest shall be conducted and if the test specimens, after immersion in water at a temperature of 75° to 85°F to a depth of 2 inches for 2 hours and tested while still wet, meet the requirements of 3.5.8, the cloth shall be considered as having passed the water resistance requirement.

TABLE III. Height for test of water resistance

Untreated weight (oz/sq yd)	Hydrostatic height		Time of exposure, minutes
	Inches	Centimeters	
Over 20	20	50.80	10
14 thru 20	14	35.56	10
Under 14	8	20.32	10

3.5.9 Flame resistance. The time of flaming of the specimens shall not exceed 2 seconds when tested as specified in 4.4. The average length of char shall not exceed the requirements of table IV. The cloth shall conform to the same fire-resistance requirements after the accelerated weathering period as specified in 4.4 and also after being exposed to water leaching when tested as specified in 4.4.

TABLE IV. Average length of char

Untreated weight (oz/sq yd)	Maximum average length of char (inches)
Under 10	4.5
10 thru 20	3.5
Over 20	2.0

3.5.10 Mildew resistance.

3.5.10.1 For types I and II cloths. The cloth shall be made mildew resistant by proper deposition within the cloth of a fungicide which is currently approved for this use by the procuring activity. Unless otherwise specified (see 6.2), the fungicide shall be one of the following in the specified concentration:

(a) Copper 8-quinolinolate to deposit a minimum of 0.18 percent copper as metal from copper 8-quinolinolate to a maximum of 0.27 percent copper as metal from copper 8-quinolinolate.

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(b) Copper naphthenate combined with copper 8-quinolinolate to deposit a minimum of 0.35 percent total copper with a minimum of 0.05 percent copper as metal from copper 8-quinolinolate to a maximum of 1.48 percent total copper with a maximum of 0.18 percent copper as metal from copper 8-quinolinolate.

The fungicide shall be uniformly applied and the cloth shall be well penetrated.

3.5.11 Breaking strength after accelerated weathering. The requirements for breaking strength after accelerated weathering shall be applicable for civil agencies procurements only.

3.5.11.1 Army and numbered duck. The breaking strength of the finished cloth after exposure to 100 hours accelerated weathering as specified in 4.4 shall conform to the following requirements:

Untreated (grey weight) (ounces per square yard)	Minimum percent retention of finished cloth breaking strength
Under 15 ounces	60.0 percent
15 ounces and over	75.0 percent

3.5.11.2 Flat ducks. The breaking strength for flat ducks CCC-C-443, after accelerated weathering, shall be not less than 80 percent of the finished cloth, or not less than that specified in the base cloth specification after exposure to 200 hours accelerated weathering, when tested as specified in 4.4.

3.6 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act (see 2.1).

3.7 Length and put-up. The cloth shall be put-up on rolls in accordance with PPP-P-1135. The minimum length of pieces shall be in accordance with the applicable grey cloth specification. The maximum length of rolls, when converted to pounds shall not exceed the maximum weight specified in PPP-P-1135.

3.8 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.9 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspectons set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.1.2 Formula approval. Material submitted to the Government when no formula approval has been obtained as required in 3.3 shall not be accepted.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement document.

4.2.2 End item examination. The end item shall be examined in accordance with the provisions of 4.2.2.1 through 4.2.2.3.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 with the following exceptions: Glossary numbers 1, 2, 3, 5, 14, 26, 27 and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks shall be scored if they result in a clearly noticeable separation of the warp yarns. Each knot and slub shall be scored as a defect only when it exceeds three times the normal yarn diameter. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 40 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 60 points. If one roll

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exceeds 60 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 60 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Color not uniform, off shade, (not within established tolerance), shaded end to end, side to side, or side to center
Poor dye penetration, mottled, streaky or cloudy
Overall application of compound not uniform
Objectionable odor
Overall uncleanness
Clearly visible crystallization of mildew inhibitors
Poorly constructed - not firmly or tightly woven
Bias or bowed filling
Tackiness (sticky to the touch)
Width not within specified tolerance

4.2.2.2 Examination for length. During the yard-by-yard examination, each roll in the sample shall be examined for put-up and length in accordance with the provisions of the untreated cloth specification.

4.2.2.3 Examination for compliance with Textile Fiber Products Identification Act and Face Marking. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act
Facing marking missing from either or both ends
Face marking on wrong side

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4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine that packaging, packing, and marking comply with section 5 requirements.

4.4 End item testing. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table VI shall be followed. The physical and chemical values specified in section 3 (except where otherwise indicated) apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1/4 yard of the desized untreated cloth and 5 continuous yards of the finished cloth, all full width of the material. The sample size (number of sample units) shall be in accordance with table V except for the inhibitor content. The sample unit for testing the inhibitor content shall be 1/4 yard of the finished cloth and the sample size shall be 5 for all lots less than 22,001 units in size, and 8 for all lots over 22,000 units in size. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified, or if the lot average determination for allowable increase in weight due to treatment exceeds requirements. Unless otherwise specified, all physical tests shall be performed under standard conditions as defined in FED-STD-191. When the data in the "Number of determinations per individual sample unit" and "Results reported as" columns are not specified in table VI, they shall be as required by the referenced test method. All test reports shall contain the individual values utilized in expressing the final results.

TABLE V. Sample size

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VI. Test methods

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Presence of mercury	3.3	1/	-	-
Nonfibrous material content	3.4.1.1	2611	-	-
Compound materials	3.4.3	1/	-	-
Yarns per inch	3.5.1	5050	-	Pass or fail
Breaking strength, initial	3.5.3	5100	-	-
Finished cloth weight increase	3.5.4	5041 2/	-	-
Dye identification	3.5.5	1/	-	-
Colorfastness to accelerated weathering	3.5.5.1	5671 3/	-	-
Fastness to crocking	3.5.6	5651	-	-

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TABLE VI. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Number determinations per individual sample unit	Results reported as
Flexibility:				
Initial				
Warp	3.5.7	5202 4.5.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Filling	3.5.7	5202 4.5.1	4	Average of 4 determinations to nearest 0.001 inch pound.
After heat aging				
Warp	3.5.7	5202 4.5.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Filling	3.5.7	5202 4.5.1	4	Average of 4 determinations to nearest 0.001 inch pound.
At low temperature				
Warp	3.5.7	5202 4.5.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Filling	3.5.7	5202 4.5.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Water resistance:				
Initial	3.5.8	5516	-	-
After accelerated weathering	3.5.8	5516 4.5.2 3/	-	-

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TABLE VI. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Flame resistance:				
Initial:				
Char length				
Warp	3.5.9	5903	4	Average of 4 determinations to nearest 0.1 inch.
Filling	3.5.9	5903	4	Average of 4 determinations to nearest 0.1 inch.
Flaming time				
Warp	3.5.9	5903	4	Each determination and average of 4 determinations to nearest 1/5 second.
Filling	3.5.9	5903	4	Each determination and average of 4 determinations to nearest 1/5 second.
After accelerated weathering:				
Char length				
Warp	3.5.9	5903 4.5.2 3/	4	Average of 4 determinations to nearest 0.1 inch.
Filling	3.5.9	5903 4.5.2 3/	4	Average of 4 determinations to nearest 0.1 inch.

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TABLE VI. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
After accelerated weathering: (cont'd)				
Flaming time				
Warp	3.5.9	5903 4.5.2 3/	4	Each determination and average of 4 determinations to nearest 1/5 second.
Filling	3.5.9	5903 4.5.2 3/	4	Each determination and average of 4 determinations to nearest 1/5 second.
After water leaching:				
Char length				
Warp	3.5.9	5903 5832 4/	4	Average of 4 determinations to nearest 0.1 inch.
Filling	3.5.9	5903 5832 4/	4	Average of 4 determinations to nearest 0.1 inch.
Flaming time				
Warp	3.5.9	5903 5832 4/	4	Each determination and average of 4 determinations to nearest 1/5 second.
Filling	3.5.9	5903 5832 4/	4	Each determination and average of 4 determinations to nearest 1/5 second.

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TABLE VI. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Mildew resistance:				
Total copper analysis	3.5.10.1	2050 or 2051 5/	-	-
Fungicide identification and concentration	3.5.10.1	6/	-	-
Breaking strength after accelerated weathering	3.5.11.1 and 3.5.11.2	5100 and 4.5.2 7/	-	8/

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ All test results on the weight of the basic untreated cloth and finished cloth shall be averaged separately to determine the respective lot average weights. These values shall then be utilized in calculating the percent increase in the finished cloth weight. Results shall be calculated to the nearest 0.1 percent.

3/ The time of exposure shall be 100 standard hours.

4/ Except that the leaching period shall be 24 hours.

5/ Contractors shall indicate on their test reports which fungicides were used.

6/ The contractor shall certify that the required amount of copper as metal from copper 8-quinolinolate has been used in the treatment of the duck. The certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

7/ The time of exposure for Army and numbered duck shall be 100 hours; and for flat duck shall be 200 hours.

8/ All test results shall be averaged and this value shall be utilized in calculating the percent change from the initial treated cloth value. Results shall be calculated to the nearest 0.1 percent.

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4.5 Methods of inspection.

4.5.1 Flexibility tests. The test shall be conducted in acordance with Method 5202 with the following modifications:

Load scale reading. The load scale reading shall be taken at a 20-degree angular deflection of the specimen.

Specimens. Eight specimens, 1 inch by 2 inches, shall be cut: four with the long dimension in the warp direction, and four in the filling direction, and pressed between two 6 by 6-inch glass plates weighted with a 20-pound weight for 4 hours at standard conditions prior to testing.

4.5.1.1 After heat aging. Flexibility test specimens shall be cut from the conditioned specimens as specified in 4.5.1. However, prior to putting them under glass plates and a 20-pound weight, the specimens shall be subjected to a temperature between 200 to 205°F for 120 hours in a well ventilated oven and then allowed to cool at standard conditions for 16 hours. The heat aged and conditioned samples shall then be placed on glass plates under weight for 4 hours prior to testing.

4.5.1.2 At low temperature. Flexibility test specimens shall be prepared as specified in 4.5.1. After the required pressing for 4 hours, weight and upper glass plate shall be removed. The lower plate holding the conditioned and pressed specimens shall then be subjected to a temperature of 0 $\pm$ 5°F for not less than 1 hour and then tested at that temperature.

4.5.2 Accelerated weathering procedure.

4.5.2.1 Apparatus and procedure. The apparatus shall be in accordance with Method 5804 except as modified below (see 6.3).

4.5.2.2 Filters. The filters shall be removed when testing all types of Army and numbered ducks listed in CCC-C-419 and flat ducks of CCC-C-443.

4.5.2.3 Procedure. Two samples 9-1/2 by 28 inches, constituting the material required to complete all weathering tests, shall be cut from the sample unit. One sample shall be cut with the long dimension in the direction of the warp; the other with the long dimension in the direction of the filling. When the width of the sample unit does not permit a full 28-inch specimen to be taken in the filling direction, two 9-1/2 by 14-inch samples shall be cut. The specimens shall be exposed one above the other in the quadrant of the accelerated weathering apparatus. The specimens shall be changed from the top to bottom racks approximately every 25 hours during the exposure period (see 3.5.11). At the conclusion of the exposure period, the samples shall be removed from the apparatus

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and allowed to dry; then the breaking strength and fire resistance specimens shall be cut from the exposed material. All specimens shall be conditioned at standard conditions prior to testing. The four fire resistance specimens shall be cut from the exposed samples such that the direction being tested shall have been exposed to accelerated weathering in the vertical position. In conducting fire resistance tests on these specimens, two specimens shall be subjected to the test flame at the end which was at the top of the exposed material, and two shall be subjected to the test flame at the end which was at the bottom of the specimen. The lower edges of the fire resistance test specimens shall be trimmed, if necessary, so that a freshly cut end is exposed to the test flame.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A, B, or Commercial as specified (see 6.2).

5.1.1 Levels A and B. The cloth shall be put-up and packaged in accordance with the level A requirements of PPP-P-1135.

5.1.2 Commercial. The cloth shall be packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A and B packing. The cloth put-up and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.2.2 Commercial The cloth, put-up and packaged as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with the applicable requirements of PPP-P-1135.

6. NOTES

6.1 Intended use.

6.1.1 Types I and II.

6.1.1.1 Class 1. Cloth treated for fire, water, weather and mildew resistance for class 1 is intended for use in the manufacture of canvas covers, tents, tarpaulins, and other duck items providing protection under conditions of prolonged outdoor use.

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6.1.1.2 Class 2 Cloth treated for fire, water, weather, and mildew resistance for class 2 is intended for use as upholstery material in motorized vehicles and uses where a higher degree of resistance to crocking is required by the condition of the end use.

6.2 Ordering data. Contracting officers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2.1).
(c) When laboratory report approval is not required (see 3.2).
(d) Submission of certification as to the place of finishing when required (see 3.2).
(e) Obtaining formula approval is mandatory (see 3.3).
(f) Base cloth required (see 3.4.1).
(g) Width of cloth required (see 3.5.2).
(h) Color required if other than specified (see 3.5.5).
(i) Flexibility requirements for duck other than numbered or Army duck specified in 3.5.7.
(j) Mildew inhibitor (if other than specified in 3.5.10.1).
(k) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Accelerated aging. Where the accelerated aging apparatus specified in 4.5.2 is not available, other tests and requirements may be submitted provided data is submitted showing correlation with the accelerated weathering test of table VI.

6.4 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.5 Colors. Dark Green FWWMR Duck may be finished under the requirements for type I. Dark Blue FWWMR Duck may be finished under the requirements for type II.

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6.6 Approval of flame resistants. As a part of the approval testing of flame retardant finishes or fiber for tentage use, the finished cloth shall have a maximum of 2 seconds afterflame and shall not melt or drop flaming pieces when tested in accordance with Method 5905 of FED-STD-191. The finished material shall retain flame resistance characteristic after 1 year of exposure to natural weathering.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - NU
Air Force - 99

Review Activities

Army - MD
Navy - MC
DLA - CT

User Activities

Army - AR, AT
Air Force - 45
DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
JUSTICE-FPI
TREASURY-OAP
USDA-AFS
VA-OSS

PREPARING ACTIVITY:

Army - GL

Project No. 8305-0750

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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